

FANUC Robot **R-30*i*B ANDR-30*i*B PLUS CONTROLLER** **Learning Path Control** **OPERATOR'S MANUAL**

MAROB LPCN02141E REV B

Applies to Version 8.20 and later

This publication contains proprietary information of FANUC America Corporation furnished for customer use only. No other uses are authorized without the express written permission of FANUC America Corporation.

**FANUC America Corporation
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253**

Copyrights and Trademarks

This new publication contains proprietary information of FANUC America Corporation furnished for customer use only. No other uses are authorized without the express written permission of FANUC America Corporation.

The descriptions and specifications contained in this manual were in effect at the time this manual was approved for printing. FANUC America Corporation, hereinafter referred to as FANUC, reserves the right to discontinue models at any time or to change specifications or design without notice and without incurring obligations.

FANUC manuals present descriptions, specifications, drawings, schematics, bills of material, parts, connections and/or procedures for installing, disassembling, connecting, operating and programming FANUC products and/or systems. Such systems consist of robots, extended axes, robot controllers, application software, the KAREL® programming language, INSIGHT® vision equipment, and special tools.

FANUC recommends that only persons who have been trained in one or more approved FANUC Training Course(s) be permitted to install, operate, use, perform procedures on, repair, and/or maintain FANUC products and/or systems and their respective components. Approved training necessitates that the courses selected be relevant to the type of system installed and application performed at the customer site.



WARNING

This equipment generates, uses, and can radiate radiofrequency energy and if not installed and used in accordance with the instruction manual, may cause interference to radio communications. As temporarily permitted by regulation, it has not been tested for compliance with the limits for Class A computing devices pursuant to Subpart J and Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference. Operation of the equipment in a residential area is likely to cause interference, in which case the user, at his own expense, will be required to take whatever measure may be required to correct the interference.

FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

FANUC America Corporation
Training Department
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253
www.fanucamerica.com

For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 1-800-47-ROBOT (1-800-477-6268). International customers should call 011-1-248-377-7159.

Send your comments and suggestions about this manual to:
product.documentation@fanucamerica.com

**Copyright © 2017 by FANUC America Corporation
All Rights Reserved**

The information illustrated or contained herein is not to be reproduced, copied, downloaded, translated into another language, published in any physical or electronic format, including internet, or transmitted in whole or in part in any way without the prior written consent of FANUC America Corporation.

AccuStat®, ArcTool®, iRVision®, KAREL®, PaintTool®, PalletTool®, SOCKETS®, SpotTool®, SpotWorks®, and TorchMate® are Registered Trademarks of FANUC.

FANUC reserves all proprietary rights, including but not limited to trademark and trade name rights, in the following names:

AccuAir™, AccuCal™, AccuChop™, AccuFlow™, AccuPath™, AccuSeal™, ARC Mate™, ARC Mate Sr.™, ARC Mate System 1™, ARC Mate System 2™, ARC Mate System 3™, ARC Mate System 4™, ARC Mate System 5™, ARCWorks Pro™, AssistTool™, AutoNormal™, AutoTCP™, BellTool™, BODYWorks™, Cal Mate™, Cell Finder™, Center Finder™, Clean Wall™, DualARM™, LR Tool™, MIG Eye™, MotionParts™, MultiARM™, NoBots™, Paint Stick™, PaintPro™, PaintTool 100™, PAINTWorks™, PAINTWorks II™, PAINTWorks III™, PalletMate™, PalletMate PC™, PalletTool PC™, PayloadID™, RecipTool™, RemovalTool™, Robo Chop™, Robo Spray™, S-420i™, S-430i™, ShapeGen™, SoftFloat™, SOFT PARTS™, SpotTool+™, SR Mate™, SR ShotTool™, SureWeld™, SYSTEM R-J2 Controller™, SYSTEM R-J3 Controller™, SYSTEM R-J3iB Controller™, SYSTEM R-J3iC Controller™, SYSTEM R-30iA Controller™, SYSTEM R-30iA Mate Controller™, SYSTEM R-30iB Controller™, SYSTEM R-30iB Mate Controller™, SYSTEM R-30iB Plus Controller™, SYSTEM R-30iB Mate Plus Controller™, TCP Mate™, TorchMate™, TripleARM™, TurboMove™, visLOC™, visPRO-3D™, visTRAC™, WebServer™, WebTP™, and YagTool™.

©FANUC CORPORATION 2017

- No part of this manual may be reproduced in any form.
- All specifications and designs are subject to change without notice.

Patents

One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

4,630,567 4,639,878 4,707,647 4,708,175 4,708,580 4,942,539 4,984,745
5,238,029 5,239,739 5,272,805 5,293,107 5,293,911 5,331,264 5,367,944
5,373,221 5,421,218 5,434,489 5,644,898 5,670,202 5,696,687 5,737,218
5,823,389 5,853,027 5,887,800 5,941,679 5,959,425 5,987,726 6,059,092
6,064,168 6,070,109 6,086,294 6,122,062 6,147,323 6,204,620 6,243,621
6,253,799 6,285,920 6,313,595 6,325,302 6,345,818 6,356,807 6,360,143
6,378,190 6,385,508 6,425,177 6,477,913 6,490,369 6,518,980 6,540,104
6,541,757 6,560,513 6,569,258 6,612,449 6,703,079 6,705,361 6,726,773
6,768,078 6,845,295 6,945,483 7,149,606 7,149,606 7,211,978 7,266,422
7,399,363

FANUC CORPORATION Patent List

4,571,694 4,626,756 4,700,118 4,706,001 4,728,872 4,732,526 4,742,207
4,835,362 4,894,596 4,899,095 4,920,248 4,931,617 4,934,504 4,956,594
4,967,125 4,969,109 4,970,370 4,970,448 4,979,127 5,004,968 5,006,035
5,008,834 5,063,281 5,066,847 5,066,902 5,093,552 5,107,716 5,111,019
5,130,515 5,136,223 5,151,608 5,170,109 5,189,351 5,267,483 5,274,360
5,292,066 5,300,868 5,304,906 5,313,563 5,319,443 5,325,467 5,327,057
5,329,469 5,333,242 5,337,148 5,371,452 5,375,480 5,418,441 5,432,316
5,440,213 5,442,155 5,444,612 5,449,875 5,451,850 5,461,478 5,463,297
5,467,003 5,471,312 5,479,078 5,485,389 5,485,552 5,486,679 5,489,758
5,493,192 5,504,766 5,511,007 5,520,062 5,528,013 5,532,924 5,548,194
5,552,687 5,558,196 5,561,742 5,570,187 5,570,190 5,572,103 5,581,167
5,582,750 5,587,635 5,600,759 5,608,299 5,608,618 5,624,588 5,630,955
5,637,969 5,639,204 5,641,415 5,650,078 5,658,121 5,668,628 5,687,295
5,691,615 5,698,121 5,708,342 5,715,375 5,719,479 5,727,132 5,742,138
5,742,144 5,748,854 5,749,058 5,760,560 5,773,950 5,783,922 5,799,135
5,812,408 5,841,257 5,845,053 5,872,894 5,887,122 5,911,892 5,912,540
5,920,678 5,937,143 5,980,082 5,983,744 5,987,591 5,988,850 6,023,044
6,032,086 6,040,554 6,059,169 6,088,628 6,097,169 6,114,824 6,124,693
6,140,788 6,141,863 6,157,155 6,160,324 6,163,124 6,177,650 6,180,898
6,181,096 6,188,194 6,208,105 6,212,444 6,219,583 6,226,181 6,236,011
6,236,896 6,250,174 6,278,902 6,279,413 6,285,921 6,298,283 6,321,139
6,324,443 6,328,523 6,330,493 6,340,875 6,356,671 6,377,869 6,382,012
6,384,371 6,396,030 6,414,711 6,424,883 6,431,018 6,434,448 6,445,979
6,459,958 6,463,358 6,484,067 6,486,629 6,507,165 6,654,666 6,665,588
6,680,461 6,696,810 6,728,417 6,763,284 6,772,493 6,845,296 6,853,881
6,888,089 6,898,486 6,917,837 6,928,337 6,965,091 6,970,802 7,038,165
7,069,808 7,084,900 7,092,791 7,133,747 7,143,100 7,149,602 7,131,848
7,161,321 7,171,041 7,174,234 7,173,213 7,177,722 7,177,439 7,181,294
7,181,313 7,280,687 7,283,661 7,291,806 7,299,713 7,315,650 7,324,873
7,328,083 7,330,777 7,333,879 7,355,725 7,359,817 7,373,220 7,376,488
7,386,367 7,464,623 7,447,615 7,445,260 7,474,939 7,486,816 7,495,192
7,501,778 7,502,504 7,508,155 7,512,459 7,525,273 7,526,121

Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

TABLE OF CONTENTS

1 GENERAL	1-1
1.1 Overview	1-1
1.2 Procedure	1-1
2 HARDWARE AND SOFTWARE	2-1
2.1 Hardware	2-1
2.2 Software	2-1
3 LPC SETUP MENU	3-1
3.1 “Main Program” Item	3-2
3.2 “Start Line” Item	3-2
3.3 “End Line” Item	3-2
3.4 “Geometric Type” Item	3-2
3.5 “Geometric Data” Item	3-2
3.6 “Process Speed” Item	3-3
3.7 “Change Blend-In Speed” Item	3-3
3.8 “Path Tolerance” Item	3-3
3.9 “Orientation Control” Item	3-3
3.10 “Max. Iterations” Item	3-4
3.11 “LEARN” function key	3-4
3.12 “RESULT” function key	3-4
3.13 “APPLY” function key	3-4
3.14 Related View	3-4
4 LPC LEARNING AND PLAYBACK	4-1
4.1 Learning	4-1
4.2 Limitations	4-2
4.3 Production with LPC Playback	4-2
5 LPC BACKUP	5-1
A ALARM CODES	A-1

1 GENERAL

1.1 Overview

- Learning Path Control (LPC) is a software option that uses iterative learning techniques to improve path performance.
- LPC is an optional function.
- The LPC function requires the Learning Path Control option (A05B-2600-R816).
- When LPC is used to improve small shapes performance, Small Circle Accuracy is recommended to be installed first.
- LPC recognizes the TP program generated by ShapeGen software. LPC's setup will be completed automatically if learning targeted program is ShapeGen created TP program.

NOTE

Model specific, contact FANUC for robot support list

1.2 Procedure

To set up this function, please follow this sequence:

1. Understand necessary hardware, necessary software, and function limitations. (Detailed information in Ch.2)
2. Set up hardware and software. (Detailed information in Ch.3)
3. Perform learning procedure. (Detailed information in Ch.4)
4. Run production

2 HARDWARE AND SOFTWARE

This chapter explains necessary hardware, necessary software, and software limitations.

2.1 Hardware

LPC requires the following hardware:

- 64MB D-RAM is necessary. LPC requires around 3MB available memory in D-RAM.

2.2 Software

2.2.1 Required Software Options

- Learning Path Control (LPC) (A05B-2500-R816) can be used in the R-30iB controller with version v8.20/P12 or later.
- Using LPC to improve small shape path performance, Small Circle Accuracy (J644) is recommended to be installed and ShapeGen software utility is recommended to be used to generate small shapes TP program.

2.2.2 Limitations

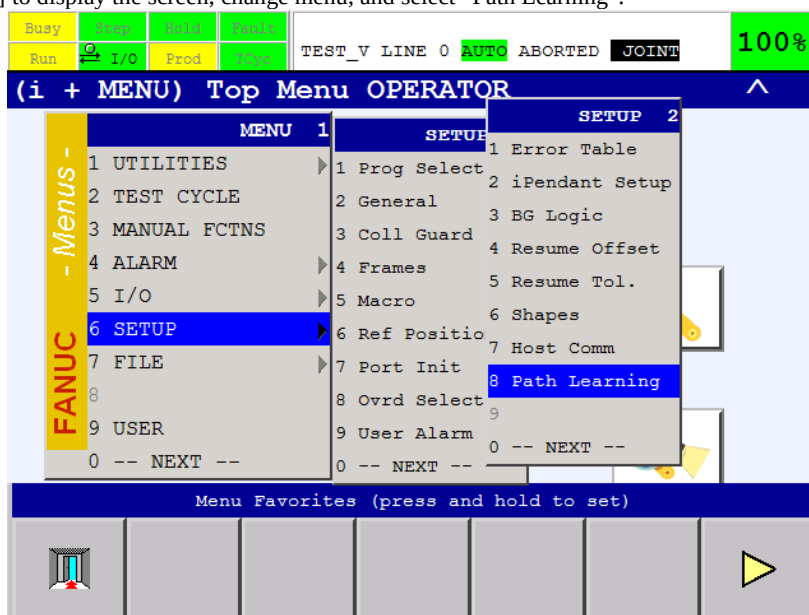
- LPC will execute targeted TP program iteratively during learning. Please make sure process such as laser or water-jet will not be turned on automatically (or is safe to be turned on) by iterative TP program execution. It may be necessary to temporarily comment out the “process on” macro before learning.
- Learning results will not be applied (path will change) under following conditions by default:
 - Single Step program execution
 - Backward program executionRobot will stop at the original taught TP position.
- LPC is currently designed and validated for circles only. Other geometric types will be supported in the future release.

3 LPC SETUP MENU

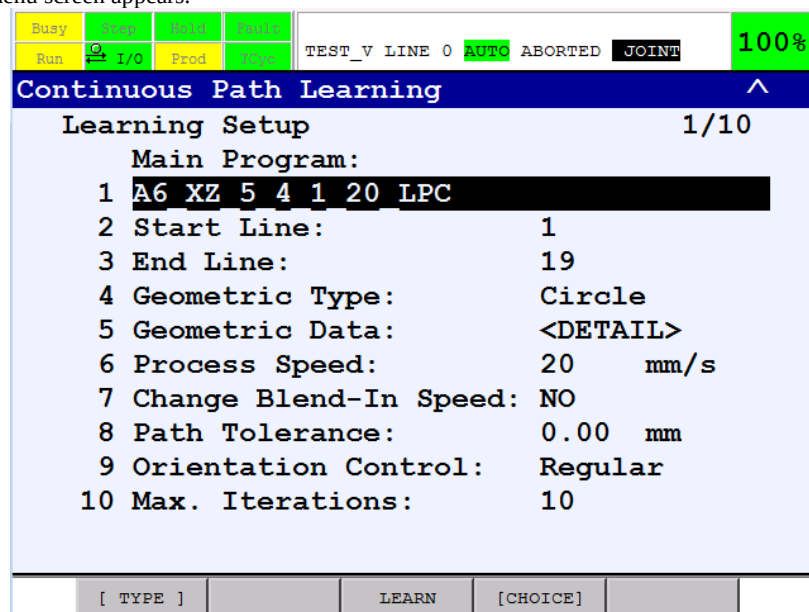
This chapter explains how to set up LPC for learning. As the first step, LPC must be configured in the setup menu. The setup menu can be launched from TP editor's related view for easy access and better display. The following procedure is used to set up LPC.

Procedure 3-1 Displaying the LPC setup menu

1. Press the MENU key.
2. Select "6 Setup".
3. Press F1, [TYPE] to display the screen, change menu, and select "Path Learning".



4. The LPC setup menu screen appears.



3.1 “Main Program” Item

This is the TP program for LPC to be executed during learning. The LRN motion modifier TP instruction will be automatically added from the Start Line to End Line to store any learned data in this targeted TP program.

If this targeted TP program is generated by ShapeGen utility, the geometric data and program related information will be extracted automatically and populate to the necessary menu items. Please double-check the setup items if the TP program was generated by ShapeGen and manually modified afterward.

3.2 “Start Line” Item

This menu item specifies the start TP line for learning process. LRN motion modifier will be added starting from this line.

3.3 “End Line” Item

This menu item specifies the end TP line for learning process. LRN motion modifier will be added before from this line.

3.4 “Geometric Type” Item

This menu item specifies the geometric type for learning process such as:

- Circle
- Slot
- Rectangle
- Free Form

Please note currently LPC will improve the path performance based on theoretic data for “Circle” type only. LPC will improve the path performance based on the taught points information for the rest of geometric types.

3.5 “Geometric Data” Item

This menu item specifies the detailed geometric type for learning process. Press Enter when cursor is at “<DETAIL>” to see the detailed data page.

3.5.1 “Geometric Data <DETAIL>” Item

The detailed geometric data for learning process is including:

- Diameter (when geometric type is Circle): the diameter of circle
- Length (when geometric type is Slot or Rectangle): the length from the origin of the shape
- Width (when geometric type is Slot or Rectangle): the width from the origin of the shape
- Center Position
- User Frame Number
- Tool Frame Number

Busy	Test	Prod	Test	TEST_V LINE 0	AUTO	ABORTED	JOINT	100%
Run	0	I/O	Prod					
Continuous Path Learning ^								
Geometric Data Setup								1/6
Geometric Type:								Circle
Diameter:								5.00 mm
Center Position:								
X: 1305.35 mm				W: -89.30 deg				
Y: 522.73 mm				P: -51.43 deg				
Z: 368.02 mm				R: -.90 deg				
User Frame Number:								0
Tool Frame Number:								1
[TYPE]								

3.6 “Process Speed” Item

This menu item specifies the programmed speed for the targeted TP program. Please note all linear and circular motion lines will be modified to this speed when learning starts.

3.7 “Change Blend-In Speed” Item

This menu item specifies whether to modify “Blend-In” motion line’s programmed speed to be the above menu item. If the targeted TP program is not generated by ShapeGen, this menu item will not take effect.

In the beginning of learning process, LPC changes programmed speed of TP lines from the specified start line to the specified end line in the targeted TP program using menu item “Process Speed” automatically. During learning iteration, the targeted TP program will be executed repeatedly and robot will move from the end of TP program back to the first line of TP program iteratively. User can use this menu item to prevent LPC changing the speed of the first line of TP program (Blend-in PT line in ShapeGen generated programs) so the learning process will be smoother and vibration free when robot travels back from the end of program.

3.8 “Path Tolerance” Item

This menu item specifies the targeted path deviation during learning. Learning process will stop once path deviation reaches this number or maximum iteration reaches the menu item. For geometric type is circle, the path deviation will calculate based on the theoretic data to the center of the circle. For other geometric types, the path deviation will calculate based on the closed taught points.

3.9 “Orientation Control” Item

When LPC is used for small shapes, J1 may be fixed to reduce the backlash impact to the path performance. To use this special 5-axis motion for small shape to ensure the perpendicular cutting wall, please select “J1 Fixed” orientation control for learning process. LRNR-005 may be posted if this TP program’s Tool orientation is not suitable for this special orientation control. Please simply change this menu item back to “Regular” and start the learning process, or spin about the tool’s approach vector (jog robot using + or - in Tool Frame Z rotation direction) and re-generate the TP program with new tool approach vector then start the learning process.

When using “J1 Fixed” orientation control, it is also recommend to use linear motion TP lines instead of circular motion TP lines. To let LPC automatically change all circular motion lines from start to end to linear motion lines, please set \$PTHLRN_CFG.\$FLAG to 2.

3.10 “Max. Iterations” Item

This menu item specifies the maximum learning iterations. Learning process will stop once path deviation reaches “Path Tolerance” menu item or maximum iteration reaches this number.

3.11 “LEARN” function key

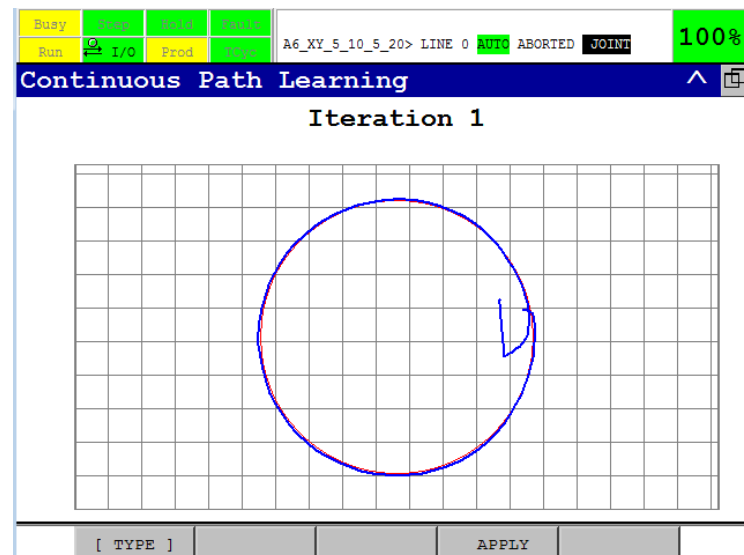
The “LEARN” function key begins the learning sequence.

- Please make sure robot is safe to run this TP program iteratively (is safe to run from the end of TP program back to the first line of TP program).
- System is in Auto mode and the speed override is at 100% and no fault present.
- When learning process starts, major keys will not be functional but any alarm/fault will stop the learning process automatically
- Learning data will store in LRN motion modifier from start to end TP lines automatically.
- If learning menu is launched from related view, the TCP trace will be displayed in lower right pane for reach iteration automatically.
- When learning stops without errors, the final maximum path deviation will be displayed.

3.12 “RESULT” function key

When learning process is complete, “RESULT” function key will show up. This function key will display the TCP trace of learning iterations. If the geometric type is circle, the theoretic circle will also display as in red for better comparison.

- Use Right or Left arrow key in this window to change the display of next or previous iteration respectively.



3.13 “APPLY” function key

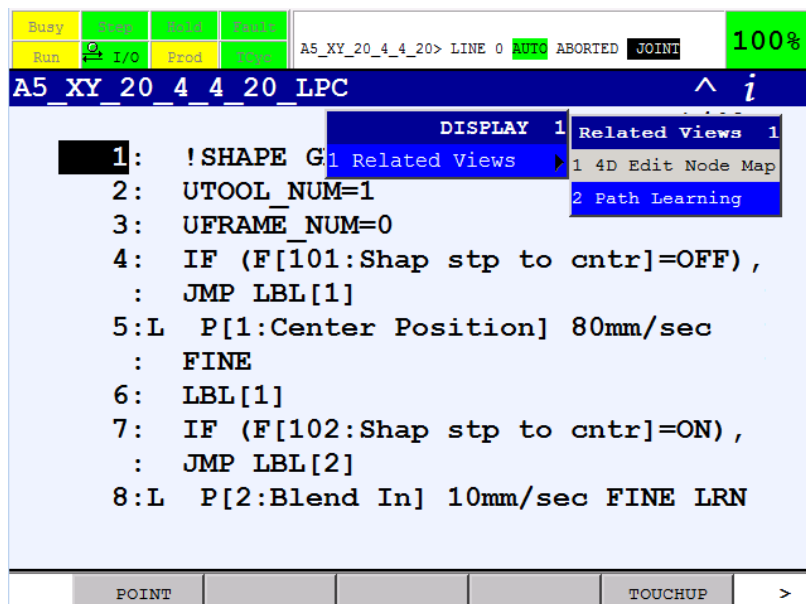
In the display window, “APPLY” function key will be available to use the displayed TCP trace as a final result and overwrite current learned data.

3.14 Related View

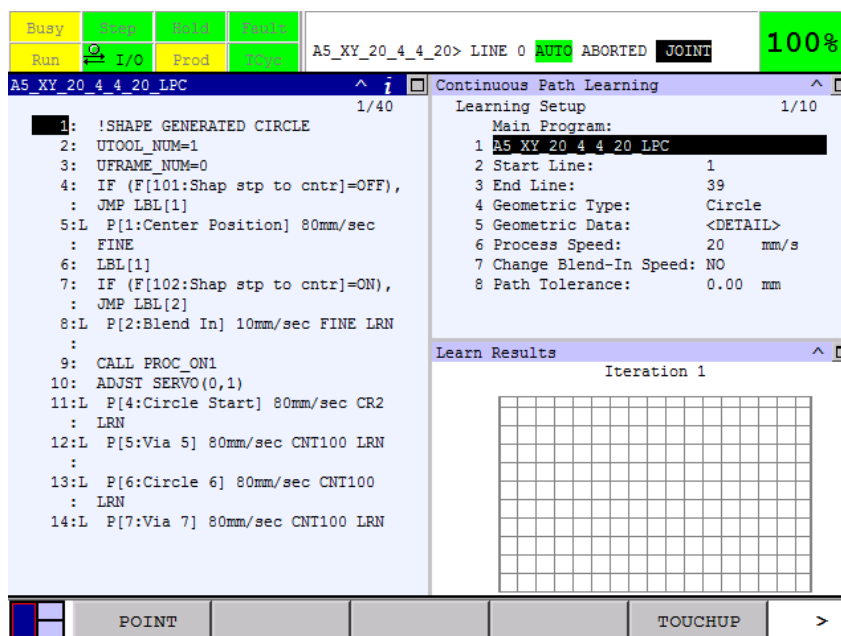
The setup menu can be launched easily when targeted TP program is opened in TP editor using related view. The TCP trace will display in the lower right pane real time during learning.

Procedure 3-2 Displaying the LPC related view menu

1. Open targeted TP program in editor.
2. Touch *i* key to bring up related view menu or press *i* + Fctn key.
3. Select “Path Learning” in related view menu.



4. After Path Learning related view windows opened, move the focus to upper right pane for setup and LEARN function key



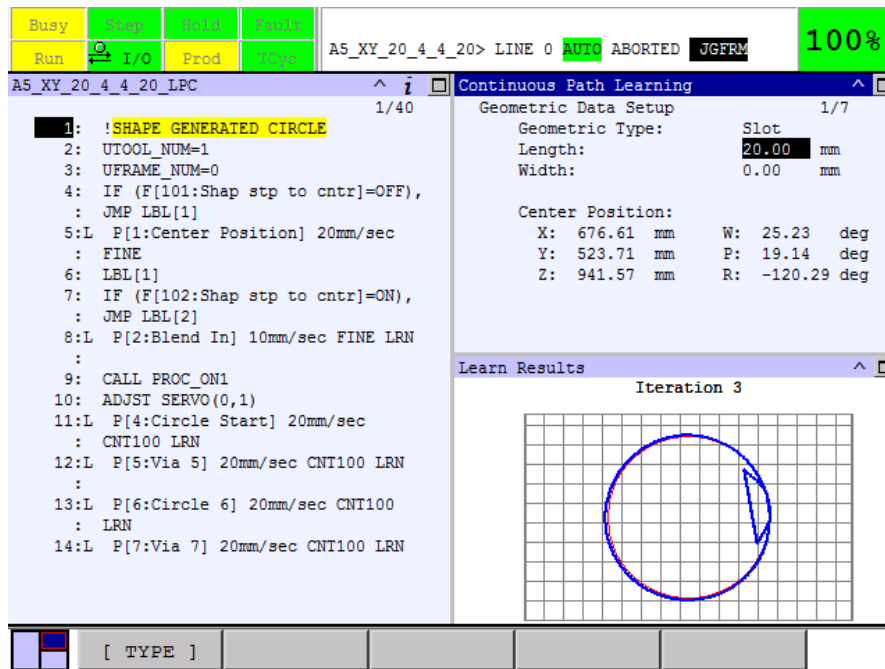
4 LPC LEARNING AND PLAYBACK

This chapter explains LPC learning and playback.

4.1 Learning

Learning will start when LEARN function key is pressed and all required conditions matched. The targeted TP program will be executed repeatedly until either path tolerance or maximum iteration is reached. During learning process, most keys on the teach pendant may not be functional. Learning can be stopped at any time when alarm occurs.

If learning is started from the related view, the TCP trace for iteration will be displayed automatically at lower right pane.



After learning is complete, the learned data is stored in LRN motion modifier for playback during production.

4.2 Limitations

- Please make sure process such as laser or water-jet will not be turned on automatically (or is safe to be turned on) by iterative TP program execution. It may be necessary to temporary comment out the “process on” macro before learning.

4.3 Production with LPC Playback

The TP program with LRN motion modifier TP instruction will apply the learned data automatically when executed. Learned data will not be applied under following conditions:

- Single step mode
- Backward execution

Robot will stop at the original taught position in above conditions.

5 LPC BACKUP

The learned data is stored in the LRN TP instructions.

The regular TP program backup procedure will backup learned data (LRN TP instruction) automatically. However, ASCII format of LS file will not have learned data converted. Please do not backup TP program after LPC learning in LS (ASCII) format.



Caution

Converting TP program with LRN motion modifier inside to LS file will lose learned data.

A

ALARM CODES

This chapter contains warnings and alarms that may occur while using LPC.

LRNR-000 Path Deviation Excess

Cause:

The TCP trace collected during learning is too far away from geometric data or taught points.

Remedy:

Make sure geometric information such as diameter/width/length, center position, utool/uframe are correct.

LRNR-001 Too many motion options

Cause:

TP line has too many motion modifiers so LRN TP instruction cannot be added.

Remedy:

Remove unnecessary motion modifiers.

LRNR-002 Learning Iteration %d ...

Cause:

The display shows learning progress.

Remedy:

Information only.

LRNR-003 Invalid learning data L:%d

Cause:

LRN data in this line number stored invalid data required large position/orientation offsets.

Remedy:

Program may have changed after learning. Re-learn this program.

LRNR-004 TCP trace mismatched

Cause:

TCP trace collected during learning is invalid.

Remedy:

Stop other TCP trace display such as 4D graphics TCP trace..

LRNR-005 Fixed J1 not reachable L:%d

Cause: Current orientation cannot be used for 5-axis motion.

Remedy:

Rotate tool in tool frame R (about approach vector) and regenerate TP program or touch TP points.

LRNR-006 Cannot record pos with LRN

Cause:

Current TP line has LRN data existed so cannot touchup this position.

Remedy:

Remove LRN motion modifier and re-learn this program if touchup is necessary.

LRNR-007 Pos type not supported L:%d

Cause:

Position representation type is not supported in this line.

Remedy:

Change the position representation type to Cartesian.

LRNR-008 Motn option not supported L:%d

Cause:

Other motion options such INC or WJNT added to the motion line with LRN existed.

Remedy:

Remove unsupported motion modifiers such as INC or WJNT.

LRNR-009 Not enough memory for learning

Cause:

Not enough memory for data collection during learning.

Remedy:

Use larger DRAM or try to use learn portion of TP program one at a time.

LRNR-010 Cannot apply learning data

Cause:

The learned data stored in LRN doesn't match with the original TP program.

Remedy:

Re-learn TP program if program has been changed.